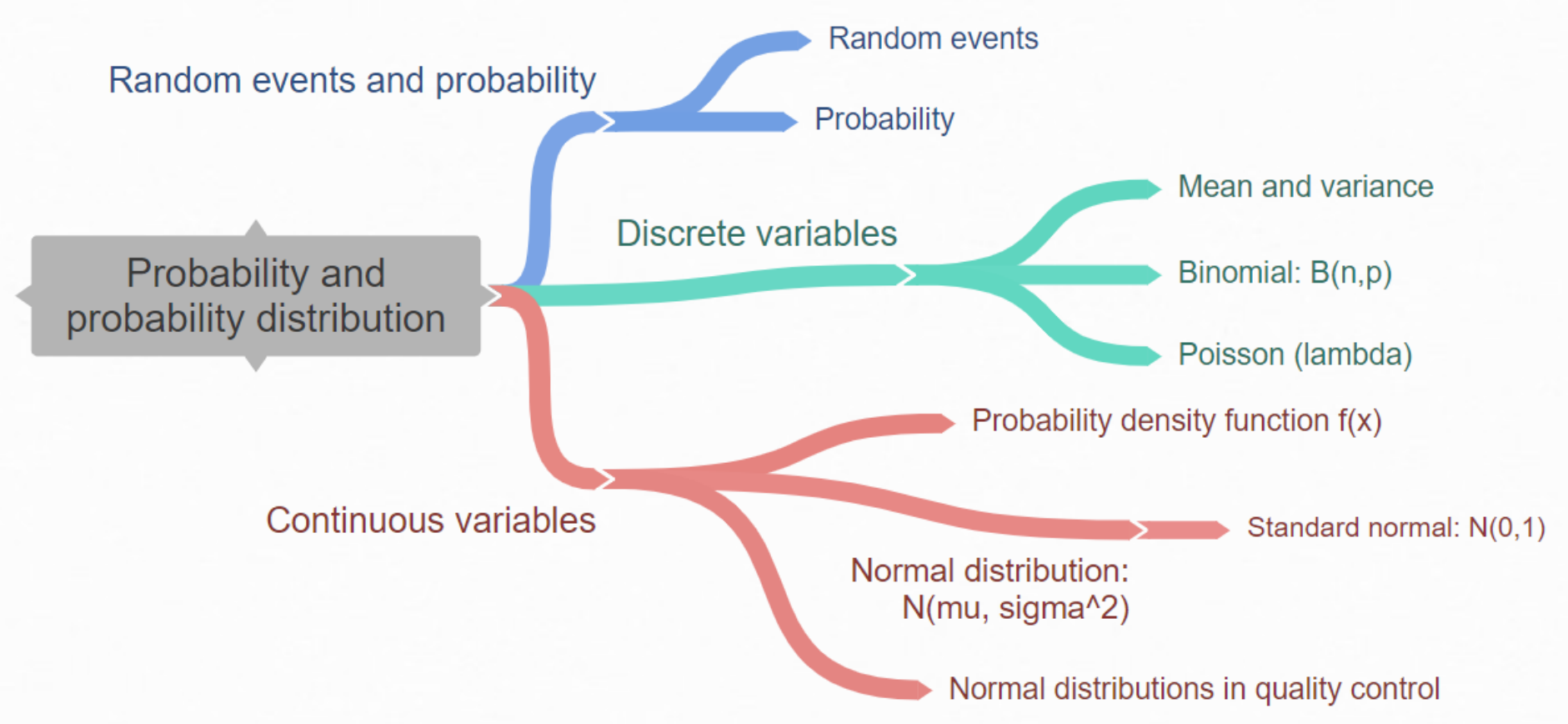
Parameter Estimation

Jiayun Jin

Review


Probability and probability distribution
Random events and probability
Random events
Probability
Discrete variables
Mean and variance
Binomial: B(n,p)
Poisson (lambda)
Continuous variables
Probability density function f(x)
Standard normal: N(0,1)
Normal distribution: N(mu, sigma^2)
Normal distributions in quality control

1. 几个重要的连续型概率分布
2. 参数和统计量
3. 中心极限定理
4. 点估计
5. 区间估计
   - 一个总体参数的区间估计
   - 两个总体参数的区间估计
   - 样本量的确定

几个重要的连续型概率分布—χ^2 分布

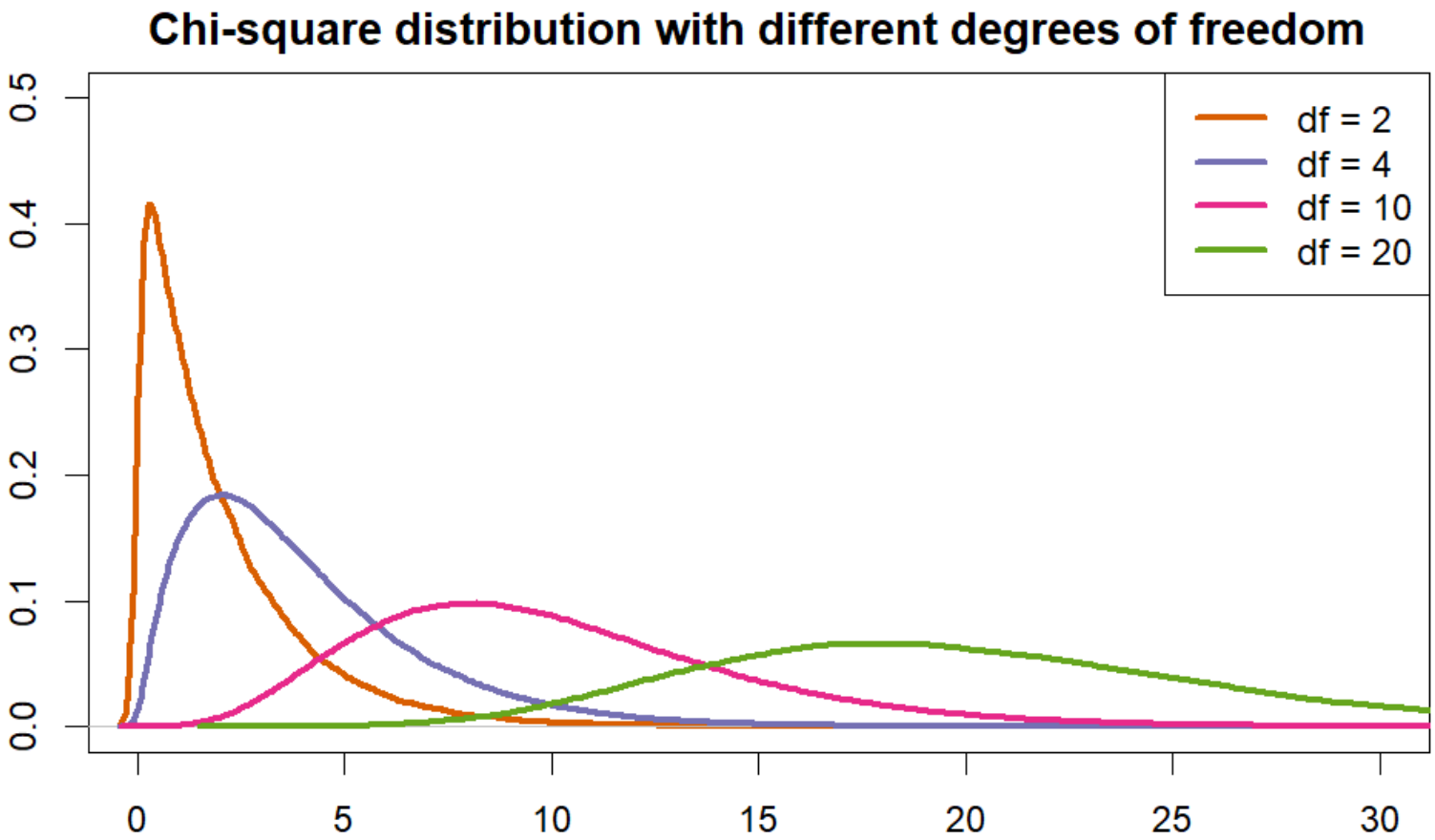
χ^2 分布

设随机变量 $X_1, X_2, ..., X_n$ 相互独立，且 $X_i (i = 1, 2, ..., n)$ 服从标准正态分布 $\mathcal{N}(0,1)$，则它们的平方和 $\sum_{i=1}^{n} X_i^2$ 服从自由度为 n 的 χ^2 分布。

$$E(\chi^2) = n$$

$$D(\chi^2) = 2n$$

χ^2 Distribution



χ^2

Properties

1. χ^2 distributions are a family of continuous probability distributions
2. Degree of freedom: n
3. The main purpose of χ^2 distributions is hypothesis testing, not describing real-world distributions.
4. The shape changes as the degrees of freedom increase: As n increases, the distribution looks more and more similar to a normal distribution.
5. Chi-square distributions start at zero and continue to infinity
6. Because the χ^2 distribution is right-skewed, the mean is greater than the median and mode.

几个重要的连续型概率分布—t 分布

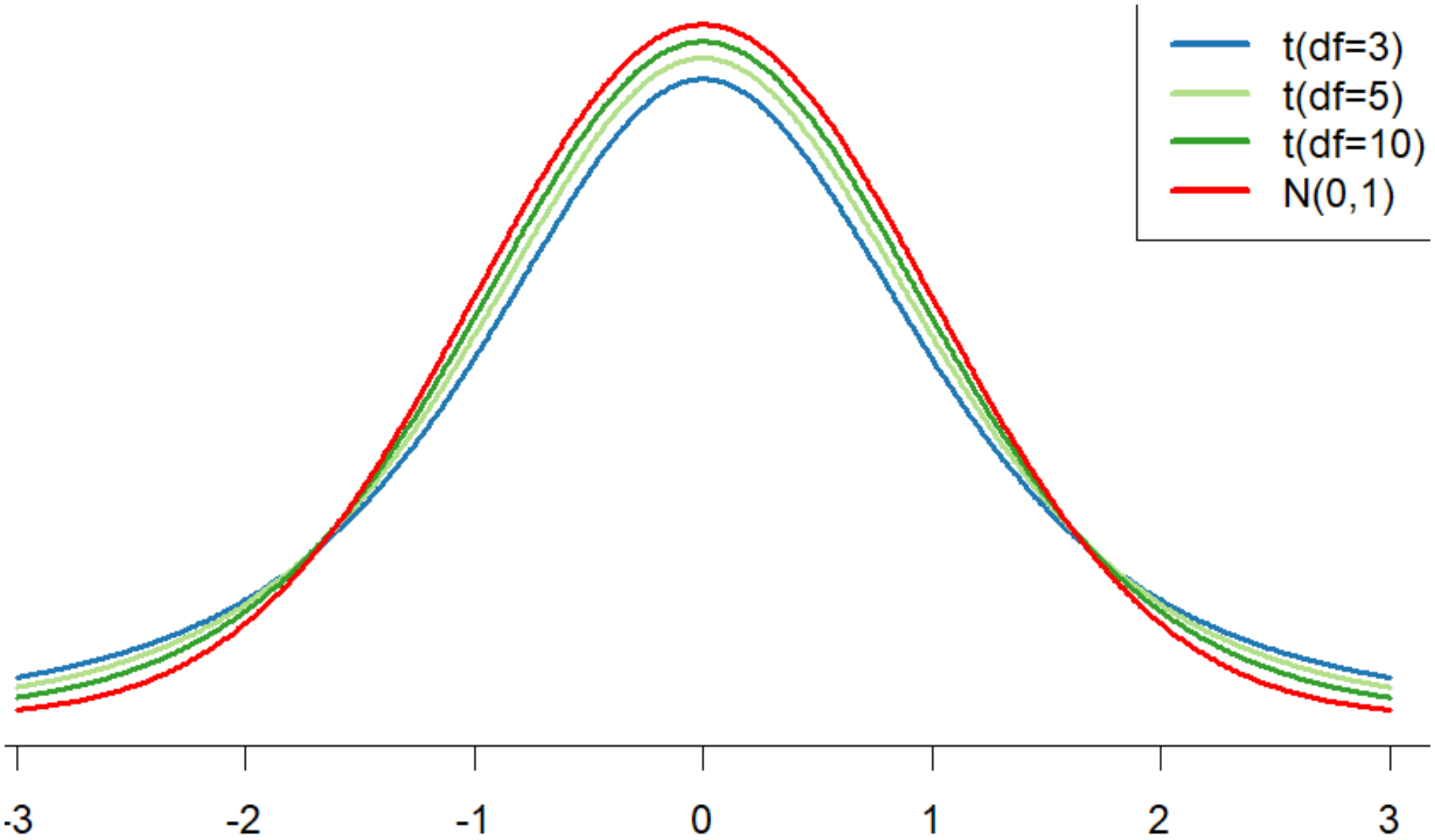
t-distribution (Student's t-distribution)

设随机变量 $X \sim \mathcal{N}(0,1), Y \sim \chi^2(n)$，且 X 与 Y 独立，则：

$$t = \frac{X}{\sqrt{Y/n}}$$

其分布称为 t 分布，记为 t(n)，其中 n 为自由度。

t distribution



t-distribution

Properties

1. A way of describing data that follow a bell curve when plotted on a graph, with the greatest number of observations close to the mean and fewer observations in the tails
2. A type of normal distribution used for smaller sample sizes
3. More conservative: It gives a lower probability to the center and a higher probability to the tails than the standard normal distribution.
4. Above 30 degrees of freedom, the t-distribution roughly matches the z-distribution.

几个重要的连续型概率分布—$\mathcal{F}$ 分布

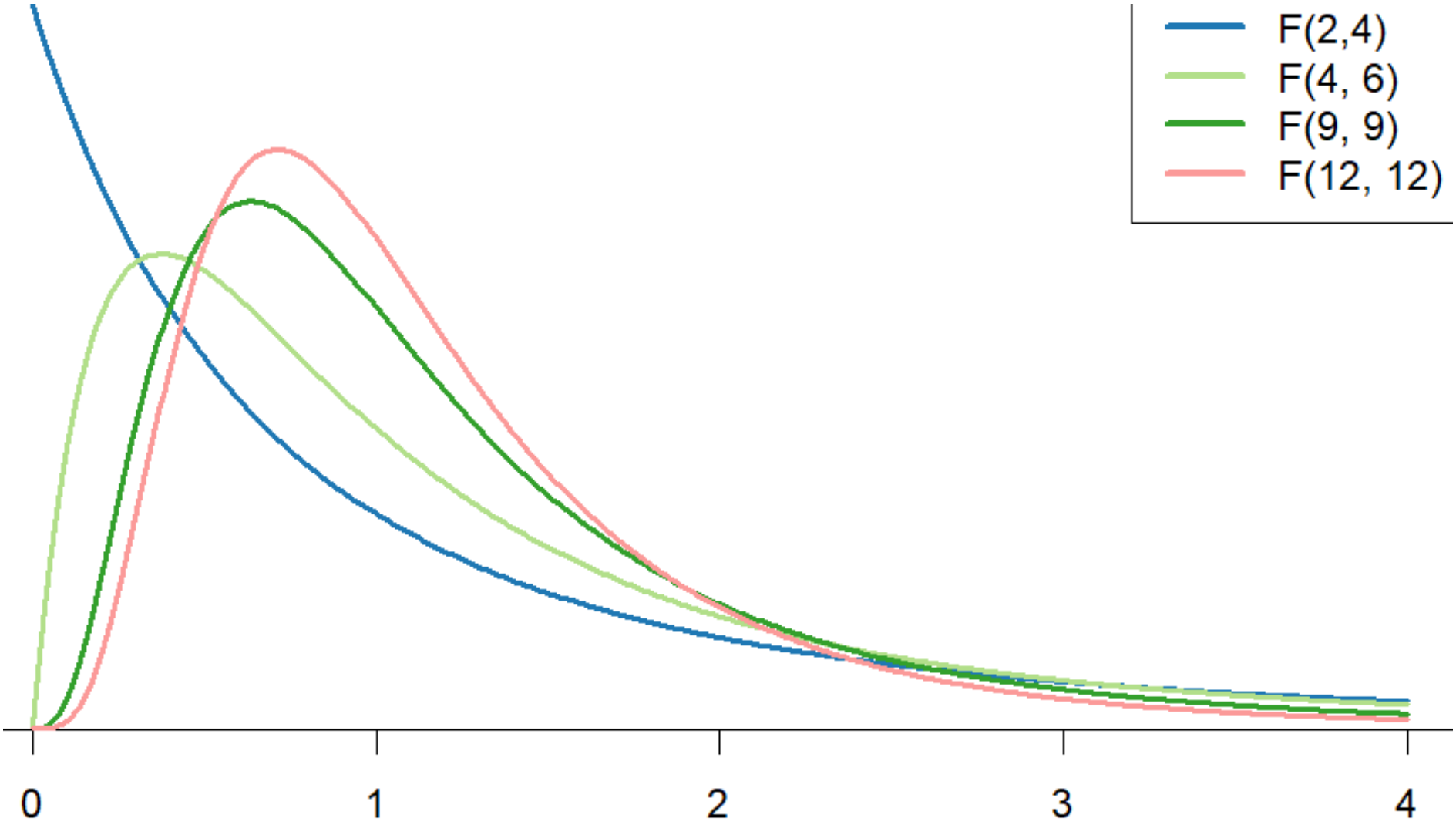
$\mathcal{F}$ 分布

设随机变量 F 与 Z 分别服从自由度为 m 和 n 的 χ^2 分布，随机变量 X 由如下表达：

$$X = \frac{Y/m}{Z/n} = \frac{nY}{mZ}$$

记为 $X \sim F(m, n)$

$\mathcal{F}$ distribution



Properties

1. The $\mathcal{F}$ is only defined for nonnegative values
2. Not symmetric: $\mathcal{F}$ distributions are generally skewed. The shape depends on the values of m and n.
3. If $X \sim F(m, n)$, $\frac{1}{X} \sim F(n, m)$

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参数和统计量

总体和样本

- 总体是指研究对象的全部个体集合
- 样本则是从总体中抽取的一部分个体集合

参数和统计量

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总体参数

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- 总体方差 σ^2

样本统计量

- 样本均值 $\bar{X}$
- 样本方差 s^2

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样本均值的分布

An experiment

1. Suppose that we draw a random sample from a population and calculate the mean from the sample
2. Now we draw another random sample of the sample size, and again calculate the mean
3. We repeat this process many times, and end up with a large number of means, one for each sample

What is the distribution of the sample means?

样本均值的分布

总体服从正态分布 $\mathcal{N}(\mu, \sigma^2)$时，随机样本 $X_1, X_2, ..., X_n$ 的样本均值

$$\bar{X} \sim \mathcal{N}(\mu, \frac{\sigma^2}{n})$$

- $\bar{X}$ 无偏性
- n 越大，$\frac{\sigma^2}{n}$ 越小

Central Limit Theorem 中心极限定理

从均值为 μ、方差为 σ^2（有限）的任意总体中抽取样本量为 n 的样本，当n 充分大时：

$$\bar{X} \sim \mathcal{N}(\mu, \frac{\sigma^2}{n})$$

- Something interesting to read:
  `http://mfviz.com/central-limit/`

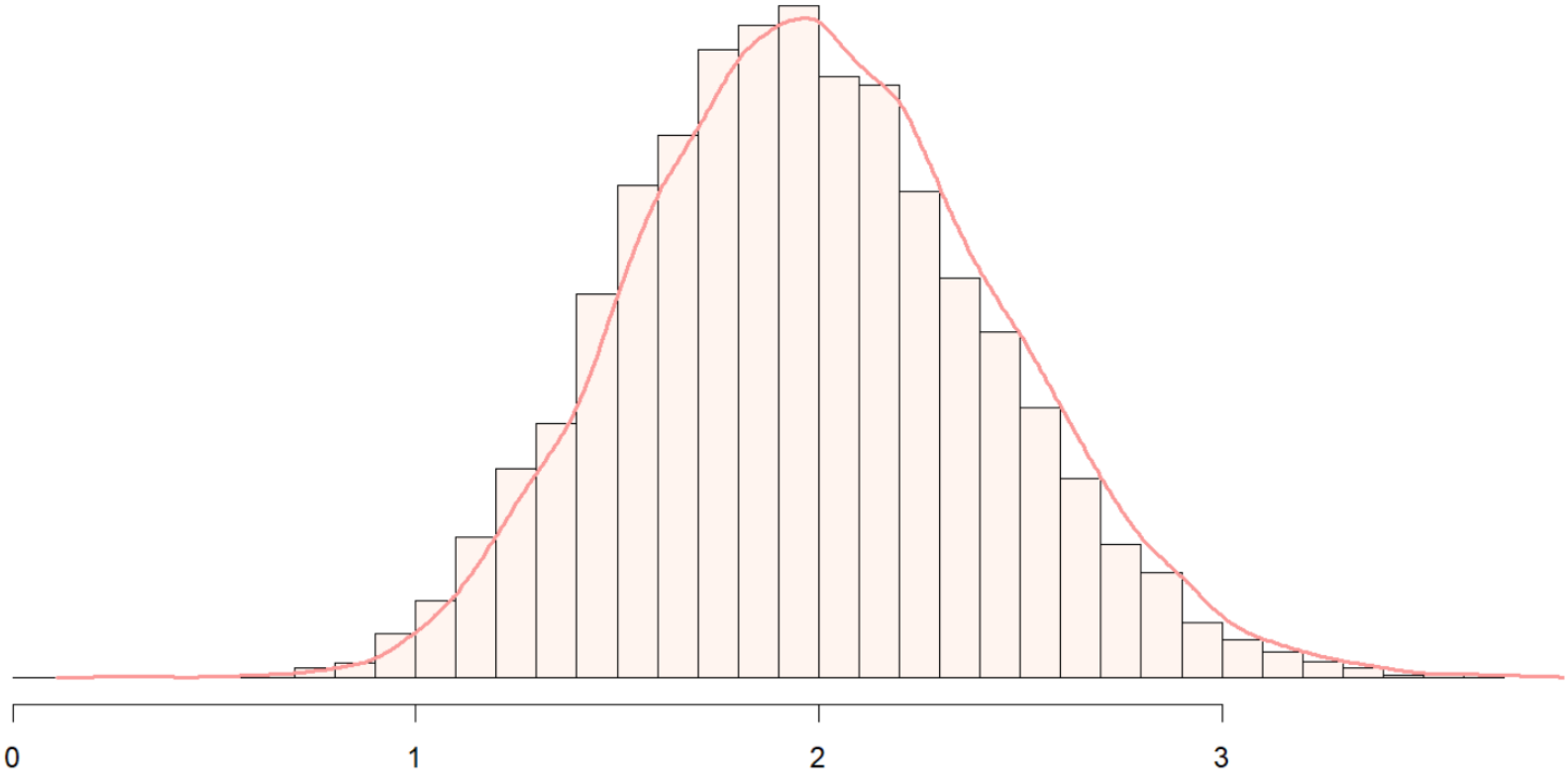
Note

- 如果样本容量足够大，不论总体分布如何，样本平均值的分布将趋近于正态分布

Central Limit Theorem

An experiment

Suppose that $X \sim poisson(\lambda = 2)$, if we take 10000 samples from the population, each with a sample size of 10, what distribution would $\bar{X}$ follow?



描述统计 vs. 推断统计

描述统计 Descriptive statistics 只关于样本

- In descriptive statistics, there is no uncertainty: the statistics precisely describe the data that you collected.
- Central tendency, variability, and the distribution

推断统计 Inferential statistics 从样本到总体

- Inferential statistics use the representative sample to make estimates and test hypothesis about the whole population
- There is uncertainty: sampling error
- Statistics $\rightarrow$ Parameter: in a way that takes sampling error into account

Inferential statistics

1. Making estimates about populations
2. Testing hypotheses to draw conclusions about population

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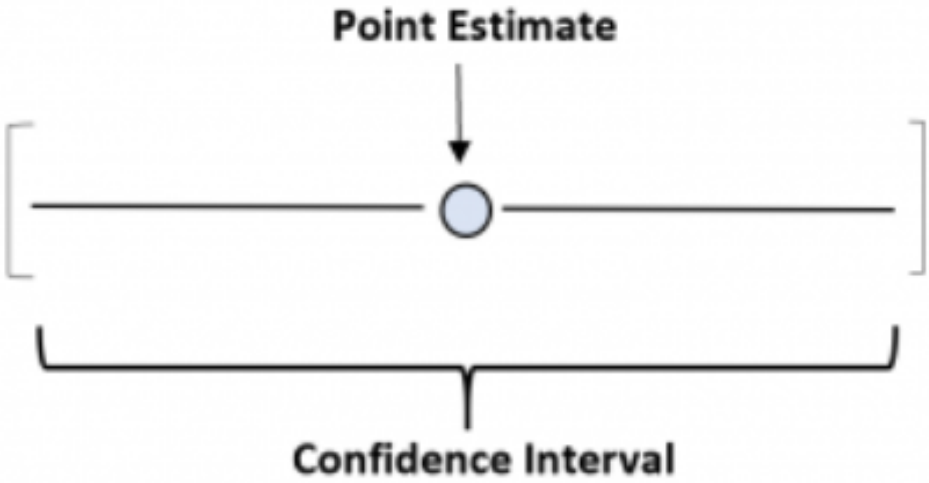
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点估计

Estimates about the population

1. **点估计 point estimation**: a single value estimate of a parameter. E.g.: $\mu = \bar{x}, \sigma = s, \pi = p$
2. **Interval estimate**: a range of values where the parameter is expected to lie. The most common type of interval estimate is a confidence interval.



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Inferential statistics

Confidence interval

1. Samples are randomly selected, so are the confidence intervals. Different samples lead to different confidence intervals.

Inferential statistics

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Inferential statistics

Confidence interval

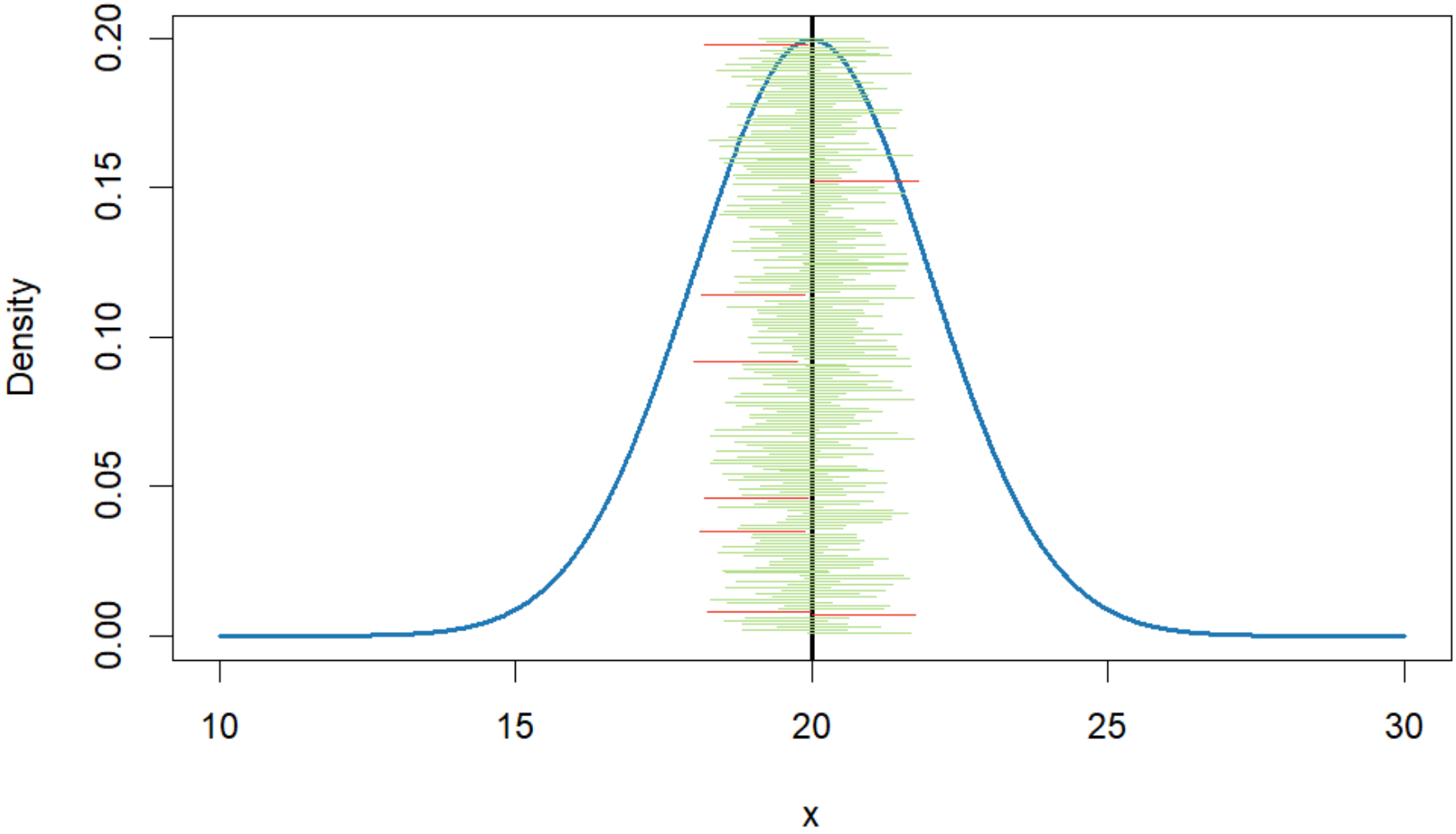
1. Samples are randomly selected, so are the confidence intervals. Different samples lead to different confidence intervals.
2. Confidence level, e.g., 95%: if we repeat with a new sample in exactly the same way 100 times, the estimate is expected to lie within the specified range of values 95 times.
3. One truth: In practice, normally we only draw one sample, and estimate one confidence interval. It either covers the population parameter or not.

Confidence interval

An example

- We want to know the average number of paid vacation days that employees at an international company receive. We collect survey responses from a random sample, and calculate a point estimate and a confidence interval
- E.g., the 95% confidence interval is calculated as [18, 24]. The unknown but existing population mean is actually 20.

Confidence interval



Inferential statistics

Confidence interval

- One population
  - Population mean: μ
  - Population proportion: π
  - Population variance: σ^2
- Two population
  - Difference between population means: $\mu_1 - \mu_2$
  - Difference between population proportions: $\pi_1 - \pi_2$
  - Variation among groups: $\frac{\sigma_1^2}{\sigma_2^2}$

Inferential statistics

To calculate a confidence interval

1. Point estimate
2. Critical values for the test statistic
3. Standard deviation of the sample
4. Sample size

Inferential statistics

Standard deviation vs standard error

- sd 标准差: describes variability within a single sample. A descriptive statistic that can be calculated from sample data.
- se 标准误: estimates the variability across multiple samples of a population. An inferential statistic that can only be estimated.

Inferential statistics

Standard deviation vs standard error

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Standard error

- se is related to sampling error. A sample will never perfectly match the population it comes from in terms of measures like means and standard deviations.
- A high se indicates that the sample may not closely represent the population

Confidence interval for one population

正态总体、方差已知, 或非正态总体、大样本

$$\bar{x} \sim \mathcal{N}(\mu, \sigma^2/n)$$

$$\frac{\bar{x}-\mu}{\sigma/\sqrt{n}} \sim \mathcal{N}(0,1)$$

the confidence interval for population mean μ with a $1-\alpha$ confidence level is:

$$\bar{x} \pm Z_{\frac{\alpha}{2}} \frac{\sigma}{\sqrt{n}}$$

where α is significance level, $1-\alpha$ is confidence level. $Z_{\frac{\alpha}{2}}$ is the critical value with $P(z \geq z_{\frac{\alpha}{2}}) = \frac{\alpha}{2}$

Note:

- $Z_{\frac{\alpha}{2}}$: 正态分布曲线下右侧面积为 $\alpha/2$ 时 z 的值
- In R: $z_{\frac{\alpha}{2}}$ $qnorm(1-\frac{\alpha}{2})$

Confidence interval for one population

正态总体、方差未知、小样本

When we use the sample variance to replace the population variance in a small dataset (typically $n \leq 30$), we use t test instead:

$$\frac{\bar{x} - \mu}{s/\sqrt{n}} \sim t(n-1)$$

the confidence interval for μ with $1-\alpha$ confidence interval is:

$$\bar{x} \pm t_{\frac{\alpha}{2}} \frac{s}{\sqrt{n}}$$

where $P(t \geq t_{\frac{\alpha}{2}}) = \frac{\alpha}{2}$

- $t_{\frac{\alpha}{2}}$ 为 t 分布曲线下右侧面积为 $\alpha/2$ 时的 t 值
- In R: $t_{\frac{\alpha}{2}} = qt(1 - \frac{\alpha}{2}, n-1)$

Summary

Population	Sample size	σ known	σ unknown
Normal	Big sample ($n \geq 30$)	$\bar{x} \pm z_{\alpha/2}\frac{\sigma}{\sqrt{n}}$	$\bar{x} \pm z_{\alpha/2}\frac{s}{\sqrt{n}}$
	small sample ($n < 30$)	$\bar{x} \pm z_{\alpha/2}\frac{\sigma}{\sqrt{n}}$	$\bar{x} \pm t_{\alpha/2}\frac{s}{\sqrt{n}}$
Non-Normal	Big sample ($n \geq 30$)	$\bar{x} \pm z_{\alpha/2}\frac{\sigma}{\sqrt{n}}$	$\bar{x} \pm z_{\alpha/2}\frac{s}{\sqrt{n}}$

Confidence interval for one population

Variance

- The χ^2 test for variance:

$$\frac{(n-1)s^2}{\sigma^2} \sim \chi^2(n-1)$$

- and then:

$$\chi^2_{1-\alpha/2} \leq \frac{(n-1)s^2}{\sigma^2} \leq \chi^2_{\alpha/2}$$

$$\frac{(n-1)s^2}{\chi^2_{\alpha/2}} \leq \sigma^2 \leq \frac{(n-1)s^2}{\chi^2_{1-\alpha/2}}$$

s^2 is sample variance, and σ^2 is population variance

Confidence interval for two populations

Difference between means

- Independent, big sample:

$$\bar{x}_1 - \bar{x}_2 \sim \mathcal{N}(\mu_1 - \mu_2, \frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2})$$

$$\frac{\bar{x}_1 - \bar{x}_2 - (\mu_1 - \mu_2)}{\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}} \sim \mathcal{N}(0, 1)$$

- The confidence interval for the difference between the means $\mu_1 - \mu_2$ with $1 - \alpha$ confidence level is:

$$(\bar{x}_1 - \bar{x}_2) \pm z_{\alpha/2} \sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}$$

Confidence interval for two populations

Variance ratio

- The ratio of two variances:

$$\frac{s_1^2\sigma_2^2}{s_2^2\sigma_1^2} \sim F(n_1 - 1, n_2 - 1)$$

$$F_{1-\alpha/2} \leq \frac{s_1^2\sigma_2^2}{s_2^2\sigma_1^2} \leq F_{\alpha/2}$$

- The confidence interval for the ratio of two variances σ_1^2/σ_2^2 with $1 - \alpha$ confidence level is:

$$\frac{s_1^2/s_2^2}{F_{\alpha/2}} \leq \frac{\sigma_1^2}{\sigma_2^2} \leq \frac{s_1^2/s_2^2}{F_{1-\alpha/2}}$$

Sample size

When estimating the population mean

- The confidence interval for the population mean:

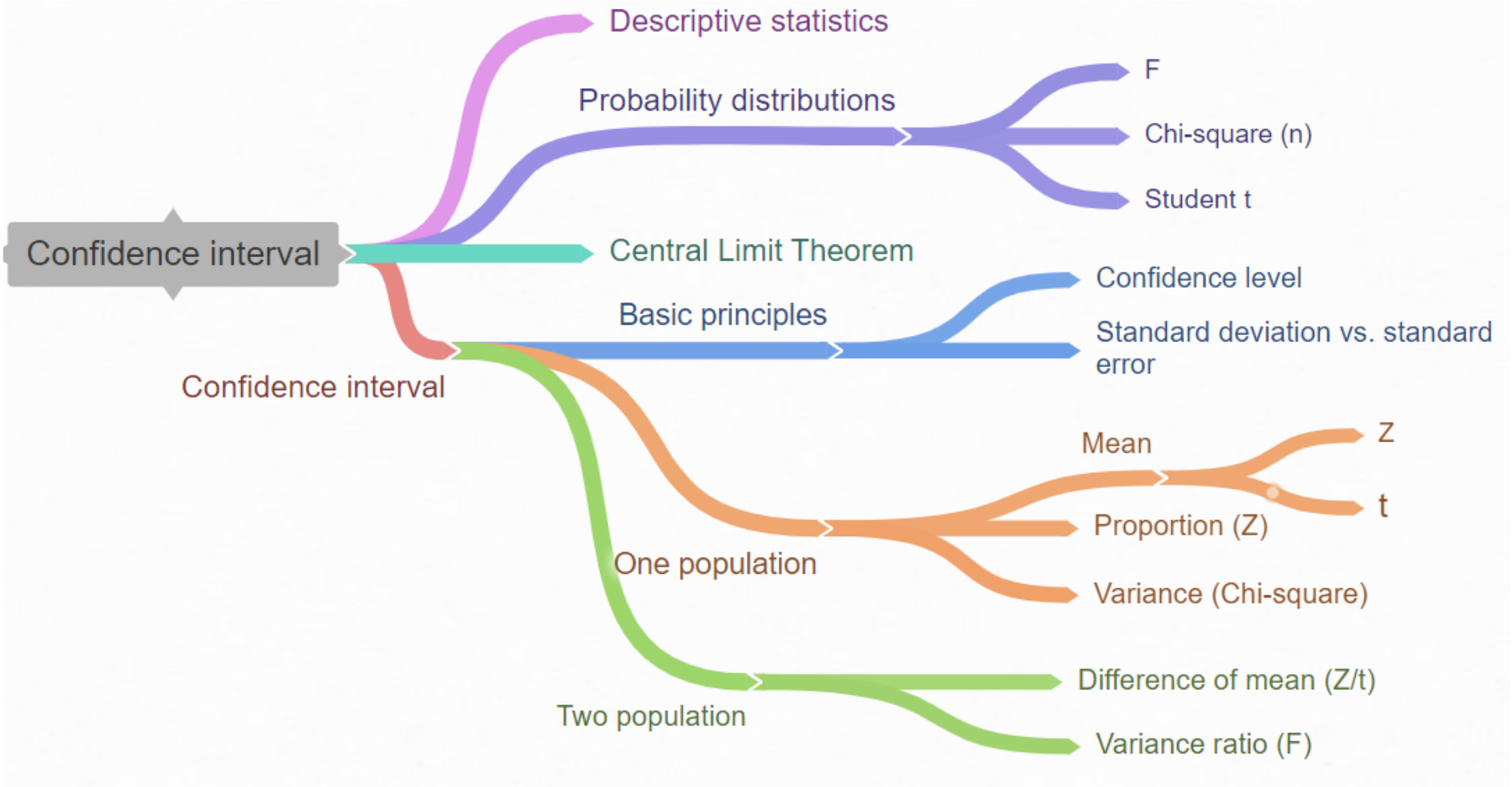
$$\bar{x} \pm z_{\frac{\alpha}{2}} \frac{\sigma}{\sqrt{n}}$$

- $$E = z_{\frac{\alpha}{2}} \frac{\sigma}{\sqrt{n}}$$

$$n = \frac{z_{\frac{\alpha}{2}}^2 \sigma^2}{E^2}$$

- With other conditions unchanged, n needs to be larger if:
  - σ increases
  - $1 - \alpha$ increases ($z_{\frac{\alpha}{2}}$ increases)
  - E decreases

Summary


Confidence interval
Descriptive statistics
Probability distributions
F
Chi-square (n)
Student t
Central Limit Theorem
Confidence interval
Basic principles
Confidence level
Standard deviation vs. standard error
One population
Mean
Z
t
Proportion (Z)
Variance (Chi-square)
Two population
Difference of mean (Z/t)
Variance ratio (F)